

Course Title: Microprocessors and Assembly Language Lab (CSE-4504)

Department of Computer Science and Engineering (CSE)
Islamic University of Technology (IUT), Gazipur

Lab # 06

Understanding Macro using Assembly Language Program.

Objective:

To understand 8086 instructions related to Macro using Assembly Language Program.

Theory:

- **Macro**

A macro is a symbolic name given to one or more assembly language statements. A macro may be used to generate instructions or data definitions.

Syntax for Declaring MACRO

```
macro_name    MACRO d1, d2, . . . , dn
               statement1
               statement2
               .
               .
               .
               statementk
ENDM
```

Where d1, d2, . . . ,dn is an optional list of dummy parameters. Dummy parameters are temporary variables; they are not declared by data definition directives (DB, DW). They can be used as input as well as output parameters.

A macro definition can appear anywhere in an assembly language program before the END directive. It is usual to place all macro definitions at the beginning of a program before the segment definitions.

The difference between *macro* and *procedure* is that *procedure* requires separate memory location to store, where as *macro* uses the same memory location of the code where the macro actually is being called. Furthermore, in *macro* parameter values can be passed.

Assembly Language Program Example for Macro and Procedure:

```
MDSPLY_STRING MACRO STRING ; Declaration of MACRO
                MOV DX, OFFSET STRING
                CALL DSPLY_STRING ; Calling the Procedure
ENDM

ORG 0100H

.DATA

MESSAGE1 DB 'Microprocessors and', 0DH, 0AH, '$'
MESSAGE2 DB 'Assembly Language$'

.CODE

MAIN PROC

MOV AX, @DATA
MOV DS, AX

MDSPLY_STRING MESSAGE1 ; 1st Call of the MACRO
MDSPLY_STRING MESSAGE2 ; 2nd Call of the MACRO

MOV AH, 4CH ; Return to DOS
INT 21H

MAIN ENDP

DSPLY_STRING PROC ; Declaration of PROCEDURE
    MOV AH, 09H
    INT 21H
    RET
DSPLY_STRING ENDP

END MAIN
```

Tasks to do:

1. Write an Assembly Language code that takes an input ARRAY of 6-decimal digits (i.e., **array DB 6 Dup (?)**) and passes the array values and address to a MACRO (i.e., **macarray MACRO oddeven**). Using the array, address and one **procedure** separate out the ODD digits and EVEN digits.

Sample Input / Output:

Input: 2, 0, 4, 7, 1, 9

Output: ODD Digits: 7 1 9
EVEN Digits: 2 0 4